

Eulerian Graphs and Digraphs

Lemma 1

If G is a graph in which the degree of each vertex is at least two, then G contains a cycle.

Proof 1. If G is a simple graph, let v_0 be any vertex of G . We construct a walk $v_0 \rightarrow v_1 \rightarrow \dots$ inductively, by choosing v_{i+1} to be any vertex adjacent to v_i ; the existence of this vertex is guaranteed by our hypothesis. Since G has only finitely many vertices, we must eventually choose a vertex that has been chosen before, and the walk between its two occurrences is a cycle. $\square$

Proof 2 (sketch). G is not a tree. $\square$

Remark. The notes leave this alternative argument as a brief hint, without further elaboration.

Theorem 2 (Euler, 1736)

A connected graph is Eulerian $\iff$ the degree of each vertex is even.

Proof. ($\Rightarrow$) Suppose that P is an Eulerian trail in G . Whenever P passes through a vertex, there is a contribution of 2 towards the degree of that vertex ($\rightarrow v \rightarrow$); since each edge occurs exactly once in P , each vertex must have even degree.

($\Leftarrow$) The proof is by induction on the number of edges of G . Suppose that the degree of each vertex is even. Since G is connected, each vertex has degree at least 2, and so by Lemma 1, G contains a cycle C . If C contains every edge of G , the proof is complete. If not, we remove from G the edges of C , to form a possibly disconnected graph H with fewer edges than G , in which each vertex still has even degree. By induction, each component of H has an Eulerian trail. Since each component of H has at least one vertex in common with C , by connectedness we obtain the required Eulerian trail of G by following the edges of C until a non-isolated vertex of H is reached, tracing the Eulerian trail of the component of H that contains that vertex, and then continuing along the edges of C ; the whole process terminates when we reach the initial vertex. $\square$

This proof can easily be modified to prove the following results.

Corollary 1

A connected graph is Eulerian $\iff$ it can be split (as a set of edges) into disjoint cycles.

Remark. No separate proof was given in the original notes; the modification of the proof of Theorem 2 is indicated but not written out.

Corollary 2

A connected graph is semi-Eulerian $\iff$ it has exactly two vertices of odd degree.

Proof. ($\Rightarrow$) Assume we have a trail T from u to v . If we add an edge between v and u , we obtain an Eulerian graph (all vertices have even degree); if we remove that edge, only u and v have odd degree.

($\Leftarrow$) We add an edge between these two vertices; then all vertices have even degree, so it is an Eulerian graph. $\square$

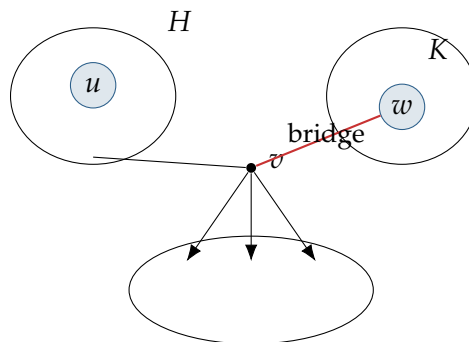
Theorem 3 (Fleury's Algorithm)

Let G be an Eulerian graph. Then the following construction is always possible and produces an Eulerian trail of G . Start at any vertex u and traverse the edges in any way, subject to this rule:

1. Erase each edge as it is traversed, and if any isolated vertex results, erase it too.
2. At each stage, use a bridge only if there exists no alternative.

Proof. We first show that the construction can be carried out at each stage. Suppose that we have just reached a vertex v . If $v \neq u$, then the subgraph H that remains is connected and contains only two vertices of odd degree, u and v . To show that the construction can be carried out, we must show that the removal of the next edge doesn't disconnect H , or equivalently, that v is incident with at most one bridge.

But if this is not the case, then there exists a bridge vw such that the component K of $H - vw$ containing w does not contain u . Since the vertex w has odd degree in K , some other vertex of K must also have odd degree, giving the required contradiction.



If $v = u$, the proof is almost identical, as long as there are still edges incident with u .

It remains only to show that this construction always yields a complete Eulerian trail. But this is clear, since there are no edges of G remaining untraversed when the last edge incident to u is used: otherwise, the removal of some smaller edge adjacent to one of these edges would have disconnected the graph, contradicting rule 2. $\square$

Theorem 4

A strongly connected digraph D is Eulerian $\iff$ for each vertex v of D , $\text{outdeg}(v) = \text{indeg}(v)$.

Proof. ($\Rightarrow$) Suppose D has an Eulerian trail. Each time the trail enters a vertex v , it uses one incoming arc; each time it leaves v , it uses one outgoing arc. Since every arc appears exactly once in the trail, $\text{outdeg}(v) = \text{indeg}(v)$.

($\Leftarrow$) We proceed by induction on the number of arcs; the result is trivial for no arcs. Assume the theorem holds for all connected digraphs with fewer arcs than D . Since D is connected and every vertex satisfies $\text{outdeg}(v) = \text{indeg}(v)$, every vertex has at least one outgoing arc.

Starting from any vertex and repeatedly following outgoing arcs, because the graph is finite, we must eventually repeat a vertex. Hence D contains a directed cycle C .

If C contains all arcs, the proof is complete. Otherwise, we remove the arcs of C from D , obtaining a digraph, possibly disconnected, H . Each component of H has fewer arcs than D , so it has an Eulerian trail. Moreover, since D is connected, every component of H shares at least one vertex with the cycle C .

We now construct an Eulerian trail of D : start traversing the directed cycle C ; when a vertex is reached that belongs to H with unused arcs, traverse the Eulerian trail of that component of H , then continue along C ; the resulting trail is a directed Eulerian trail of D . $\square$